MXR M87 Bass Compressor

Contents

1. Description

MXR M87 Bass Compressor is a compressor pedal for bass guitar.

About compression

In audio engineering, the compression effect allows you to dynamically control the difference between the loudest and quietest parts of an audio signal.

In terms of bass guitar, a dedicated compressor effect can:

- Improve the volume consistency of played notes.
- Increase the sustain of the notes.
- Make the quieter nuances more audible in the mix.
- Control the sharp peaks from slaps and pops when playing with the slapping technique.
- Help you shape the distorted tone by raising certain frequencies more.

In general, all compressors have at least four basic controls:

Threshold

The volume above which the compression begins.

Ratio

The strength of the compression; the higher the ratio, the more dB is reduced from the input above the threshold.

Attack

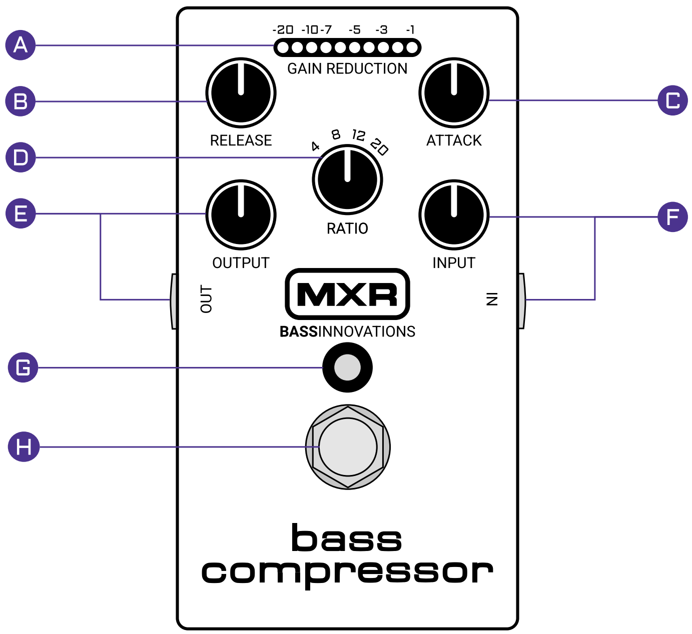
The time it takes for the compressor to start reducing the signal after it exceeds the threshold.

Release

The time the compression lasts after it initially reducing the signal.

2. Controls

Use the pedal's control knobs to shape how your signal is compressed.



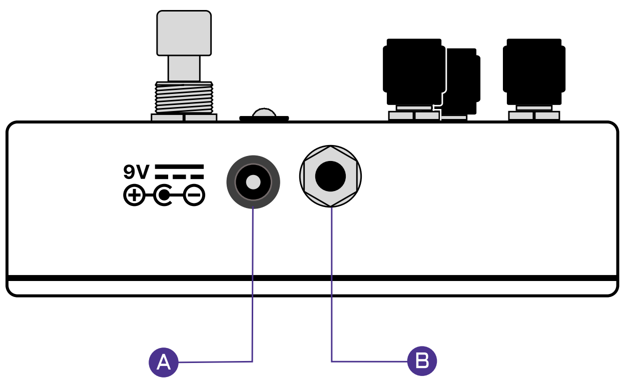
A	The **GAIN REDUCTION** meter shows the level of compression applied in real time. If it doesn't light up, it means that the input signal has not reached the fixed threshold for compression.
B	The **RELEASE** knob controls how quickly compression stops after the signal falls back down.
C	The **ATTACK** knob controls how quickly compression is applied after a note is played.
D	The **RATIO** determines how strongly the signal is compressed once it exceeds the threshold.
E	The **OUTPUT** knob controls the overall volume of the signal coming out of the pedal. The **OUT** 1/4" jack on the side of the pedal connects to an amplifier or other pedals with an instrument cable.
F	The **INPUT** knob controls the gain of initial signal coming out of your guitar. The **IN** 1/4" jack on the side of the pedal connects to the guitar with an instrument cable.
G	The light indicator shows whether the pedal is on or off.
H	The footswitch toggles the pedal on or off.

3. Power

The pedal requires 9V DC power to operate.

You can power the pedal in two ways:

- A standard 9V battery. Remove the bottom lid of the pedal with a screwdriver to install the battery.
- A 9V AC/DC adapter. Connect it either to a standard AC wall socket or a compatible power supply. Plug the power connector into the power jack next to the **IN** jack for the instrument.



A	9V DC center-negative power jack.
B	1/4" jack for the instrument cable. Labeled **IN** on the front plate.

4. First start

This is a basic workflow with the compressor as your only pedal.

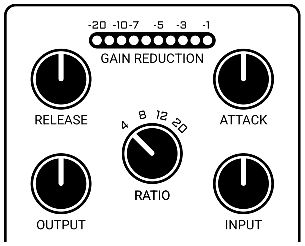
Before you begin, make sure the pedal has power *(on page 3)*.

Starting from basic settings, try to experiment with all the knobs to get an effect you enjoy.

1. Plug in the cable from your guitar into the **IN** jack input.

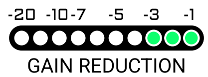
2. Run a different cable from the pedal's **OUT** jack to your amplifier or audio interface.

3. You can start off with the following settings on the pedal:
    a. **RELEASE**, **ATTACK**, **OUTPUT**, and **INPUT** knobs at 12 o'clock.
    b. **RATIO** at 4 o'clock.



4. Press the footswitch to turn on the pedal.

5. Looking at the **GAIN REDUCTION** meter, rotate the **INPUT** knob so that you see a range from 3 to 7 lights on the meter while playing your guitar normally:



6. Rotate the **OUTPUT** knob to the desired overall volume.
7. Rotate the **ATTACK** knob clockwise to soften the initial attack of your note or counterclockwise to accentuate it.
8. Rotate the **RELEASE** knob clockwise to make the compression effect last longer or counterclockwise to make it fade faster.
9. Use the **RATIO** knob to define the strength of compression. The bigger the setting (up to 20:1), the more volume will be decreased when compression is in effect. For example, the 4:1 setting means that, for every 4dB over the threshold, only 1dB comes through.

Once you have dialed in the compression, you can start playing as-is or add other effects after the compressor in your signal chain.

5. Best practices

Things to keep in mind when using compression.

Ratios

- Start with a more gentle compression ratio (4:1) to simply even out your dynamics while keeping the natural tone and punch of your bass.
- The higher your ratio, the more compressed your sound will become, affecting the tone.
- A 20:1 ratio is effectively a hard limiter and 12:1 is a soft limiter. For a "brick wall" effect, use limiters with fast attack and release settings. A slow release with a fast attack will give you a "breathing" effect. Lower the input to only apply limiters to the loudest peaks of the signal (like slapping).

Input and threshold

- When dialing in the **INPUT** knob, observe the **GAIN REDUCTION** meter. It shows how much the input signal is hitting the threshold.
- If you want compression to apply only to the loudest peaks of your playing without affecting what you consider the normal range, make sure the **GAIN REDUCTION** meter does not go further than 3 LEDs while playing normally. In this case, the meter should only reach above 3 LEDs when you make your loudest sounds. You might have to raise the ouput volume to compensate for the lower input.

Other tips

- To get the best results with the highly distorted tones, split the bass signal by sending the low end into a compressor and the high end into a distortion. This requires separate gear to split by frequencies.
- If you use an EQ, it's generally better to use it after the compressor in the signal chain.
- If you are a beginner, you might want to practice without a compressor to develop your natural dynamics.

6. Specifications

The list of technical details of the MXR M87 Bass Compressor.

Type	Value	Description
Input Impedance	1 MΩ	
Output Impedance	600 Ω	
Max Input Level	+14 dBV	
Max Output Level	+8.5 dBV	
Frequency Response	±1 dB, 20 Hz to 20 kHz	
Noise floor	-90 dBV	A-Weighted, with all controls at the middle position.
Total harmonic distortion (THD)	> 0.5 %	20 dB gain reduction, 1.1s release setting, 50 Hz to 20 kHz.
Gain	31 dB	
Compression Ratio	4:1, 8:1, 12:1, 20:1	
Attack time	20 μs to 800 μs	
Release time	50 ms to 1.1 s	
Bypass	True Hardwire	
Current Draw (LEDs OFF)	14 mA	
Current Draw (LEDs ON)	19 mA	

Type	Value	Description
Power Supply	9V DC	